LAND & LEDGERS
Occupation Architecture, Demography & Workforce Calibration

Expanded Final Report - 112 Occupation Profiles + Population, Work, Mortality and Workforce Weights

Derived design report based on Canonical Project Bible Revision XXV, Technical Implementation Bible Revision VIII, U.S. Census occupational/population tables and historical mortality/labor references

25 September 2026

Scope lock: 112 core occupation profiles, 300+ possible historical display titles, and shared task/capability/equipment systems. The implementation tiers below are development priorities, not social rank, wage rank or historical importance.

1. Executive Summary

This expanded report keeps the final 112-profile occupation architecture, then adds the demographic and labor-force calibration needed to populate the world: age/sex structure, gainful-work participation, household/family work, career entry and soft-exit ages, occupational hazard/physical-strain classes, and a 100.000% baseline worker-weight model for all 112 occupations.

The five implementation tiers remain development-priority tiers derived from the supplied Canon and Technical Bible. The new population and occupation weights are calibration data, not new Canon locks: historical Census totals anchor the broad structure, while fine-grained 112-job weights and sex priors are explicitly model-derived bridge estimates that must be modified by settlement role, year, infrastructure and local demand.

The report continues to distinguish underlying occupation profiles from display titles, civic offices, ownership, management authority, employment terms and current assignments. It now also separates census-like gainful employment from unpaid household/family production, so women, children and older Persons can contribute real labor without being forced into a wage-occupation statistic.

2. Source Basis and Design Doctrine

Canon Revision XXV - Opening: The normal Open Frontier begins in spring 1870. The opening southeastern Dakota/Missouri-oriented economy is sparse, physically dispersed, dependent on overland and river movement, and may use the existing 55-95 Person settlement target. Productive work can occur in houses, barns, yards and mixed-use premises rather than requiring a dedicated storefront.

Canon Revision XXV - Transformation phases: 1870-72 emphasizes wagon/river logistics and thin finance; 1873-77 brings first local rail and crisis; 1878-87 brings rapid rail/town growth, machinery and early advanced-node utilities; 1887-95 has mature rail networks and financial stress; 1896-1905 has deeper institutions and selected mature communications.

Canon Revision XXV - Multimodal transport: Roads, stages, ferries, freight routes, river landings and steamboats remain first-class infrastructure. Rail changes route economics rather than deleting wagon and local freight work.

Technical Revision VIII - Farm generation: Farms are generated from component state such as land, acreage, buildings, draft animals, livestock, equipment access, household labor, hired labor, capital/debt, enterprise mix and connectivity rather than a single FarmLevel.

Technical Revision VIII - Employment: Employment should resolve through one employment relationship authority rather than independent profession/wage/workplace truths.

Technical Revision VIII - Occupation profiles: Occupation profiles map typical tasks and capabilities to requirement sets; they are data, not class gates. A Person may perform a task when capability, authority and resources resolve.

3. Implementation Tiers

Tier 1 - Foundation - Must Have First. Implement before or during the first playable 1870 economy. These profiles support the default rural settlement, essential transport, basic crafts, trade, lodging and schooling.

Tier 2 - Early Settlement - High Priority. Needed soon after the foundation so small settlements and early regional networks feel complete. These are common services, professions and non-rail transport roles, but not every 55-95 Person settlement must contain each one.

Tier 3 - Growth & Network Economy. Add for the 1870s-1880s expansion into finance, mechanization, processing, newspapers, telegraphy, grain handling and railroad service.

Tier 4 - Regional / Mature / Infrastructure-Dependent. Required for full release when the correct geography, industry, institution or mature infrastructure exists: mining, riverboats, ranching, utilities, heavy machinery, larger rail operations and similar systems.

Tier 5 - Rare / Late / Specialist. Still part of the full-release target, but safe to implement last because these profiles are low-frequency, late-period, highly specialized or scenario-specific. Tier 5 does not mean historically unimportant.

Important: a Tier 4 or Tier 5 occupation can be essential inside the correct scenario. For example, Miner is Tier 4 for default development order but becomes foundational inside a Black Hills mining scenario. Tier measures implementation priority across the default full game, not in-world prestige or local frequency.

Tier	Profiles	Share of 112
1	12	10.7%
2	27	24.1%
3	28	25.0%
4	31	27.7%
5	14	12.5%

4. Final-Final Pass: What Changed

No additional occupation was removed. The final count remains 112.

Cowboy/Range Hand is normalized as Stock Hand / Range Hand, with Cowboy, Cowhand, Herder and Drover available as display titles or assignment descriptions.

Artisan is renamed Specialty Artisan to prevent it from becoming a generic catch-all for established trades; it is reserved for low-frequency hand-production crafts that do not merit their own underlying profile.

The stop rule is now explicit: keep a separate profile when a distinction materially changes required capabilities, equipment, task logic, labor mobility or institutional responsibility; otherwise preserve the historical name through specialization, employer, office, rank or assignment.

The system must not create new occupations merely by adding an industry prefix. Railroad Carpenter, Mine Carpenter and Military Blacksmith remain Carpenter or Blacksmith/Farrier employed in different industries.

5. Master Catalog - 112 Occupation Profiles

Agriculture & Rural

#	Occupation	Tier	One-line role definition
1	**Farmer**	**1**	Operates a farm and coordinates land, crops, livestock, household labor, hired labor and equipment; specialties such as wheat, dairy or tenant farming remain farm-state data.
2	**Rancher**	**4**	Operates a primarily livestock/range enterprise whose land use, herd management and labor pattern differ materially from an ordinary mixed farm.
3	**Farm Hand**	**1**	Performs ordinary hired or family farm work; seasonal, live-in, harvest, hay and milking variants are employment terms or assignments rather than separate occupations.
4	**Stock Hand / Range Hand**	**4**	Handles livestock on range or in herds; display titles can include cowboy, cowhand, herder or drover according to the current work.
5	**Agricultural Machinery Operator**	**3**	Operates specialized farm machinery or traveling custom equipment such as threshers, reapers and steam-powered agricultural outfits.
6	**Horse Trainer**	**4**	Breaks, trains and conditions horses for riding, driving or work, usually as a specialist service rather than ordinary stable labor.
7	**Stable Hand**	**2**	Feeds, grooms, harnesses and tends horses in liveries, hotels, stage stations and other stables; hostler and groom are display variants.

Labor & Overland Transport

#	Occupation	Tier	One-line role definition
8	**Laborer**	**1**	Provides general manual labor where no specialist trade is required, including loading, excavation, road work, warehouse work and construction support.
9	**Teamster**	**1**	Drives and manages horse- or ox-drawn freight and delivery teams; drayman, freight driver and local deliveryman are assignment-based display titles.
10	**Stagecoach Driver**	**2**	Operates scheduled passenger, mail or stage service where route discipline and passenger responsibility justify a distinct occupation.
11	**Messenger**	**3**	Carries documents, telegrams, notices and small parcels locally for businesses, institutions or communications offices.
12	**Mail Carrier**	**2**	Carries official mail over assigned routes; route type, contract status and later rural delivery are service or agreement details.

Building, Repair & Craft

#	Occupation	Tier	One-line role definition
13	**Blacksmith / Farrier**	**1**	Performs forge work, tool and wagon iron repair and horseshoeing, with smithing and farriery tracked as separate capabilities within one profile.
14	**Carpenter**	**1**	Performs structural and finish wood construction, including framing, roofing and bridge carpentry when the Person has the required specialization.
15	**Cabinetmaker**	**3**	Produces and repairs furniture, cabinetry and finer shop-built woodwork that is more specialized than ordinary building carpentry.
16	**Mason**	**2**	Builds in brick or stone using mortar; bricklayer and stonemason are specializations rather than separate underlying profiles.
17	**Stonecutter**	**4**	Cuts and dresses stone for buildings, monuments and other finished work before masons place it.
18	**Plasterer**	**3**	Applies and repairs interior or exterior plaster finishes and related prepared surfaces.
19	**Painter / Glazier**	**2**	Paints and finishes structures and can cut, fit and repair window glass when qualified; sign painting is a specialization.
20	**Tinsmith**	**2**	Fabricates and repairs sheet-metal household, building and commercial goods, often

#	Occupation	Tier	One-line role definition
			overlapping with roofing and early plumbing work.
21	**Plumber**	**4**	Installs and repairs water, drainage, steam and other pipe systems as such infrastructure becomes locally supportable.
22	**Mechanic**	**3**	Repairs general machinery and equipment; farm machines, bicycles, early automobiles and similar specialties are familiarity domains.
23	**Machinist**	**4**	Performs precision metal machining, fitting and repair using machine tools beyond ordinary blacksmith work.
24	**Millwright**	**4**	Installs, aligns and maintains industrial machinery, shafts, belts and mill equipment.
25	**Boilermaker**	**4**	Builds and repairs boilers and heavy riveted pressure vessels used by steam-powered industry and transport.
26	**Wagonmaker / Wheelwright**	**1**	Builds and repairs wagons, carriages and wheels; wheelwright and carriage-maker can remain display specialties.
27	**Saddler / Harness Maker**	**1**	Makes and repairs saddles, harness, tack and related working leather goods for horse-powered transport and agriculture.
28	**Shoemaker**	**2**	Makes and repairs boots and shoes; bootmaker and cobbler are display variants.
29	**Tailor**	**2**	Cuts, fits, makes and repairs tailored clothing, especially fitted garments and men's wear.
30	**Dressmaker / Seamstress**	**1**	Makes, alters and repairs clothing and household sewn goods; millinery can be an added specialization.
31	**Jeweler / Watchmaker**	**5**	Sells, makes and repairs jewelry, watches and clocks, generally appearing only in larger or more mature commercial centers.
32	**Gunsmith**	**5**	Repairs and sometimes builds firearms and related mechanisms as a low-frequency specialist trade.
33	**Specialty Artisan**	**5**	Umbrella profile for low-frequency hand-production crafts such as cooper, brickmaker, broom maker, soap maker, candle maker, mattress maker, basket maker, bookbinder or cigar maker.

Commerce, Finance & Administration

#	Occupation	Tier	One-line role definition
34	**Merchant**	**1**	Owns or operates ordinary retail trade; hardware, grocery, dry-goods, feed, lumber, coal and similar labels come from the business's merchandise specialization.
35	**Clerk**	**2**	Performs customer, record and administrative work in stores, hotels, banks, railroads, government offices and other institutions.
36	**Bookkeeper / Accountant**	**3**	Maintains financial records and, at higher capability, performs reconciliation, statements, auditing and more complex accounting work.
37	**Banker**	**3**	Directs banking, lending and deposit activity; president, cashier and loan officer are institutional positions or authority levels within the profile.
38	**Agent**	**3**	Represents another party or arranges transactions, covering land, insurance, express, freight, livestock, auction and other brokerage or agency work.
39	**Grain Dealer**	**3**	Buys, markets and finances bulk grain, requiring commodity, quality, storage and transport knowledge beyond ordinary retailing.

Hospitality, Personal Service & Leisure

#	Occupation	Tier	One-line role definition
40	**Hospitality Keeper**	**1**	Operates lodging or meal establishments; hotel keeper, innkeeper, boarding-house keeper and eating-house keeper are business-dependent display titles.
41	**Saloon Keeper**	**2**	Operates a saloon and manages alcohol inventory, service, regulation, credit and any separately authorized entertainment or gambling.
42	**Cook**	**2**	Prepares meals for hotels, boarding houses, restaurants, camps, institutions or wealthy households.
43	**Server**	**2**	Provides table and customer service in eating houses, hotels, restaurants and similar establishments.
44	**Housekeeper**	**2**	Maintains guest rooms or larger households and may supervise cleaning, bedding and room readiness.
45	**Bartender**	**2**	Serves drinks, maintains the bar station and handles customer transactions under a saloon or hospitality operator.
46	**Gambling Dealer**	**4**	Runs period table games such as faro or poker; the bankroll owner or gaming-room manager is an ownership or authority role, not another occupation.
47	**Musician**	**4**	Performs instrumental music for dances, saloons, hotels, churches, theatres or private instruction and events.
48	**Entertainer**	**5**	Performs acting, singing, dancing or variety entertainment when these are a principal livelihood.
49	**Sex Worker**	**5**	Provides independent or establishment-based sexual services where the scenario, local law and social conditions support the occupation.
50	**Barber**	**2**	Provides hair cutting, shaving and related grooming services, usually from a small shop or shared commercial space.
51	**Domestic Servant**	**2**	Performs paid household service such as cleaning, childcare assistance, meal support and general domestic work.
52	**Laundress**	**2**	Performs paid washing and ironing for households, hotels or businesses, including independent home-based work.

Resource Extraction & Processing

#	Occupation	Tier	One-line role definition
53	**Logger**	**4**	Fells and handles timber in logging operations; hauling is performed by Teamsters while general loading can use Laborers.
54	**Sawyer**	**3**	Operates sawmill cutting equipment and converts logs into lumber, distinct from the engine operator powering the mill.
55	**Miller**	**2**	Operates grist, flour or roller-milling processes that turn grain into flour, meal or feed.
56	**Dairy Processor**	**3**	Processes milk into butter, cheese or creamery products; cheesemaker and butter maker are display specializations.
57	**Brewer**	**4**	Produces beer and related brewery output; malting and cellar work can be capabilities or subordinate labor assignments.
58	**Baker**	**2**	Produces bread, pastry and other baked goods; confectionery can be a specialization rather than a separate profile.
59	**Butcher**	**2**	Slaughters, cuts, prepares and sells meat using specialized tools and food-handling facilities.
60	**Elevator Operator**	**3**	Operates a grain elevator as a storage and transfer facility linking producers, merchants and rail shipment.
61	**Grain Weigher / Inspector**	**3**	Determines shipment weight and commercial quality or grade where applicable, directly affecting settlement and acceptance of grain.

Professional, Health & Institutional

#	Occupation	Tier	One-line role definition
62	**Physician**	**2**	Diagnoses and treats human illness and injury; surgery is a capability within the physician profile rather than a separate occupation.
63	**Nurse**	**4**	Provides skilled patient care in homes, hospitals or institutions where dedicated nursing work is supported.
64	**Midwife**	**2**	Provides childbirth and maternal care as a distinct community service that may exist outside formal medical institutions.
65	**Dentist**	**5**	Provides specialized dental treatment and prosthetic work, generally appearing in larger or more mature population centers.
66	**Veterinarian**	**4**	Diagnoses and treats animal illness and injury as a dedicated profession where livestock demand can support it.
67	**Druggist / Pharmacist**	**2**	Compounds and dispenses medicines while operating the specialized retail and record functions of a drugstore or pharmacy.
68	**Medical Attendant**	**4**	Provides non-physician, non-nurse clinical or institutional assistance such as orderly, clinic or hospital-attendant work.
69	**Lawyer**	**3**	Provides legal advice, document preparation and representation; judge, prosecutor, notary and similar authorities are offices or licenses layered on top.
70	**Teacher**	**1**	Provides formal instruction in rural schools, town schools or private settings; head teacher and superintendent are authority assignments.
71	**Clergy**	**2**	Provides ordained religious leadership and services; minister, preacher, priest and ordained missionary are institution-dependent display titles.
72	**Undertaker**	**3**	Arranges funerals and handles preparation of the deceased, with embalming as a later capability rather than a separate profile.

Print, Media & Culture

#	Occupation	Tier	One-line role definition
73	**Editor / Publisher**	**3**	Directs and publishes a newspaper or periodical, often combining ownership, editorial judgment and business management in smaller towns.
74	**Reporter**	**4**	Gathers and writes news for publications; in a small paper the editor may perform this work personally.
75	**Compositor**	**3**	Sets and arranges type for printing, a skilled trade distinct from operating the press itself.
76	**Printer**	**3**	Operates printing presses and performs commercial or newspaper printing once type and material are prepared.
77	**Photographer**	**5**	Produces photographic portraits, commercial images and documentary work using period cameras and chemical processes.
78	**Artist**	**5**	Produces visual art, illustration or commercial creative work when that is a meaningful livelihood rather than an occasional skill.

Engineering & Technical

#	Occupation	Tier	One-line role definition
79	**Professional Engineer**	**4**	Performs civil, railroad, mining, mechanical or late-period electrical engineering; this is distinct from a locomotive engineer.
80	**Architect**	**5**	Designs buildings and prepares architectural plans for clients able to support a dedicated professional service.
81	**Draftsman**	**4**	Produces technical drawings for engineering, architecture, railroads, mining and other project work.
82	**Surveyor**	**2**	Measures and records land, routes and sites for settlement, property, public land, construction and transportation projects.

#	Occupation	Tier	One-line role definition
83	**Stationary Engineer**	**3**	Operates fixed engines, boilers, pumps and industrial power machinery in mills, waterworks, mines and power plants.

Public Safety & Security

#	Occupation	Tier	One-line role definition
84	**Peace Officer**	**2**	Performs public law-enforcement work; sheriff, deputy, marshal, constable and policeman are offices or jurisdictional display titles.
85	**Detective**	**5**	Performs specialized investigative work for private clients, railroads or institutions where such demand exists.
86	**Guard**	**3**	Provides watch, security, jail or access-control work without the full public authority of a peace officer.
87	**Firefighter**	**4**	Performs organized fire response; in small towns this may be a secondary volunteer civic role rather than a person's main livelihood.
88	**Scout / Guide**	**5**	Provides route finding, terrain knowledge, tracking, guiding and reconnaissance as a paid specialty.

Railroad Operations

#	Occupation	Tier	One-line role definition
89	**Station Agent**	**3**	Runs the railroad's local station function and may combine ticketing, freight, express, telegraph and local operating authority.
90	**Track Worker**	**3**	Builds, inspects and maintains track, ties, ballast and related right-of-way infrastructure; section foreman is an authority assignment.
91	**Brakeman**	**3**	Handles train braking, switching-related duties and car movement under the conductor and operating rules of the railroad.
92	**Conductor**	**3**	Directs the operation and commercial handling of a train and coordinates crew, passengers or freight.
93	**Locomotive Engineer**	**3**	Operates a locomotive and is responsible for controlling the train's motive power and safe handling.
94	**Train Dispatcher**	**4**	Coordinates train movements over a railroad territory using schedules, orders and communications systems.
95	**Switchman**	**4**	Moves and organizes railcars within yards, sidings and station areas and operates switches as required.

River, Ferry & Steam Transport

#	Occupation	Tier	One-line role definition
96	**Riverboat Officer**	**4**	Serves as captain or mate with command and deck authority aboard a river vessel; rank is modeled separately.
97	**Riverboat Pilot**	**4**	Navigates the vessel using detailed knowledge of river channels, hazards, water conditions and landings.
98	**Riverboat Engineer**	**4**	Operates and maintains a riverboat's propulsion machinery and engine-room systems.
99	**Steam Fireman**	**3**	Fires boilers and manages fuel and steam pressure aboard locomotives, riverboats or other steam operations with domain familiarity.
100	**Deckhand**	**4**	Performs vessel deck work, lines, loading assistance and routine handling under officers aboard a riverboat.
101	**Ferryman**	**2**	Operates a local ferry crossing and handles the movement of people, animals, wagons and goods across a water barrier.

Communications & Utilities

#	Occupation	Tier	One-line role definition
102	**Telegraph Operator**	**3**	Sends, receives and records telegraph messages at an office, station or other connected node.
103	**Telephone Operator**	**5**	Operates a telephone exchange or switchboard after local telephone networks become established.
104	**Lineman**	**3**	Constructs and maintains telegraph, telephone, electrical or signal lines according to the network and the worker's specialization.
105	**Electrician**	**5**	Installs and repairs electrical equipment and wiring in the later period where electric systems are locally available.

Mining & Metallurgy

#	Occupation	Tier	One-line role definition
106	**Miner**	**4**	Extracts mineral or stone resources underground, at the surface or in quarries; prospector and quarryman can be task or display specializations.
107	**Driller**	**4**	Drills rock for mining, quarrying or blasting work using hand or powered drilling methods appropriate to the site.
108	**Powderman**	**4**	Handles and uses explosives for mining, quarrying, railroad or major construction work

#	Occupation	Tier	One-line role definition
			and therefore requires specialized competence.
109	**Assayer / Metallurgist**	**5**	Tests ores and evaluates their composition or treatment, ranging from frontier assay work to more advanced metallurgical expertise.
110	**Ore-Mill Operator**	**4**	Runs ore-crushing, stamp-milling or concentration processes that prepare or recover value from mined material.
111	**Smelter Worker**	**4**	Works in furnaces and associated processes that reduce or refine ore at a smelter.

Military & Federal

#	Occupation	Tier	One-line role definition
112	**Soldier**	**4**	Serves in the military; enlisted and officer ranks are modeled as rank and authority rather than separate occupation profiles.

6. Implementation Order by Tier

Tier 1 - Foundation - Must Have First

Farmer; Farm Hand; Laborer; Teamster; Blacksmith / Farrier; Carpenter; Wagonmaker / Wheelwright; Saddler / Harness Maker; Dressmaker / Seamstress; Merchant; Hospitality Keeper; Teacher.

Tier 2 - Early Settlement - High Priority

Stable Hand; Stagecoach Driver; Mail Carrier; Mason; Painter / Glazier; Tinsmith; Shoemaker; Tailor; Clerk; Saloon Keeper; Cook; Server; Housekeeper; Bartender; Barber; Domestic Servant; Laundress; Miller; Baker; Butcher; Physician; Midwife; Druggist / Pharmacist; Clergy; Surveyor; Peace Officer; Ferryman.

Tier 3 - Growth & Network Economy

Agricultural Machinery Operator; Messenger; Cabinetmaker; Plasterer; Mechanic; Bookkeeper / Accountant; Banker; Agent; Grain Dealer; Sawyer; Dairy Processor; Elevator Operator; Grain Weigher / Inspector; Lawyer; Undertaker; Editor / Publisher; Compositor; Printer; Stationary Engineer; Guard; Station Agent; Track Worker; Brakeman; Conductor; Locomotive Engineer; Steam Fireman; Telegraph Operator; Lineman.

Tier 4 - Regional / Mature / Infrastructure-Dependent

Rancher; Stock Hand / Range Hand; Horse Trainer; Stonecutter; Plumber; Machinist; Millwright; Boilermaker; Gambling Dealer; Musician; Logger; Brewer; Nurse; Veterinarian; Medical Attendant; Reporter; Professional Engineer; Draftsman; Firefighter; Train Dispatcher; Switchman; Riverboat Officer; Riverboat Pilot; Riverboat Engineer; Deckhand; Miner; Driller; Powderman; Ore-Mill Operator; Smelter Worker; Soldier.

Tier 5 - Rare / Late / Specialist

Jeweler / Watchmaker; Gunsmith; Specialty Artisan; Entertainer; Sex Worker; Dentist; Photographer; Artist; Architect; Detective; Scout / Guide; Telephone Operator; Electrician; Assayer / Metallurgist.

7. Historical Titles That Should NOT Become New Occupation Profiles

Historical/display title	Underlying model
Dairy Farmer / Wheat Farmer / Tenant Farmer	Farmer + enterprise/tenure state
Harvest Hand / Hay Hand / Seasonal Farm Laborer	Farm Hand + assignment/employment term
Cowboy / Drover / Cattle Herder	Stock Hand / Range Hand + current assignment
Drayman / Freight Driver / Deliveryman	Teamster + service/assignment
Hostler / Groom	Stable Hand + workplace/assignment
Livery Keeper	Qualified Person + livery ownership/management; often Stable Hand or Merchant with authority
Railroad Carpenter / Mine Carpenter	Carpenter + employer/industry
Railroad Blacksmith / Military Blacksmith	Blacksmith / Farrier + employer/industry
Hardware Merchant / Grocer / Lumber Dealer / Feed Merchant	Merchant + inventory specialization
Insurance Agent / Land Agent / Express Agent / Auctioneer	Agent + specialization/authority
Bank Cashier / Loan Officer	Banker + bank position/authority
County Sheriff / Marshal / Constable / Policeman	Peace Officer + office/jurisdiction
Mayor / Treasurer / Assessor / Recorder	Civic office layered on the Person
Judge / Prosecutor / Notary	Lawyer or qualified Person + office/license
Postmaster	Government office; often layered onto Merchant or Clerk
Federal Land Office Register / Receiver	Clerk + federal office/position

Historical/display title	Underlying model
Manager / Foreman / Superintendent	Management authority, not a standalone occupation
Owner / Proprietor	Ownership relationship, not occupation
Seasonal / Permanent / Live-in	Employment agreement terms
Family Worker	Household/employment relationship
Interpreter	Language + interpretation capability/service
Appraiser	Capability/service of a qualified domain specialist
Auditor	Bookkeeper / Accountant specialization
Embalmer	Undertaker capability
Roofer / Shingler	Carpenter or Tinsmith + roofing specialization
Milliner	Dressmaker / Seamstress + millinery specialization
Cheesemaker / Butter Maker	Dairy Processor + process specialization
Confectioner	Baker + product specialization
Railroad Signalman	Lineman or Track Worker + signal specialization
Locomotive Hostler	Rail/steam worker + roundhouse/hostling assignment
Wharfmaster	Agent or Clerk + landing authority/position
Librarian	Clerk + library position, except if a later design chooses dedicated professional librarianship
Ice Cutter / Ice Deliveryman	Laborer or Teamster + seasonal ice-harvest/delivery assignment
Well Digger	Laborer + specialty/contract work

8. Services Remain Separate from Occupations

The occupation roster should not become a proxy for the service catalog. Businesses and Persons expose services through actual capabilities, assets, inventory, agreements and infrastructure. The following service families should remain separate data even when several are delivered by one occupation or business:

Agriculture & livestock: Custom plowing/seeding/harvest/threshing; machinery hire; breeding; horse training; farriery; veterinary work; fencing; livestock handling and weighing.

Freight & passenger transport: General hauling, farm-to-town freight, drayage, local delivery, stage service, carriage service, ferry crossing, express and warehousing.

Rail: Passenger, freight, mail/express, stock shipment, grain shipment, station handling, switching, track maintenance and rolling-stock servicing.

Retail & procurement: General merchandise, specialized inventories, special orders, delivery, store credit and historically appropriate barter/trade arrangements.

Craft & repair: Smithing, horseshoeing, wagon/wheel repair, harness and shoe repair, carpentry, masonry, painting, machine repair, watch repair and specialty artisan production.

Construction & property: Contract building, excavation, framing, roofing, masonry, plastering, painting, plumbing, surveying, engineering and maintenance.

Processing: Milling, sawmilling, baking, butchering, dairy processing, brewing, grain elevation/weighing and ore processing.

Hospitality & leisure: Rooms, boarding, meals, saloon service, bathing/laundry, horse stabling, gambling, music, entertainment and scenario-appropriate vice services.

Professional: Medical, midwifery, dental, veterinary, pharmacy, legal, accounting, surveying, engineering, architectural and agency/brokerage services.

Finance: Deposits, loans, mortgages, private lending, merchant credit, note discounting, collections and insurance/brokerage relationships where historically supported.

Postal & communications: Mail carriage, telegraph messages, message delivery, telephone exchange service and network maintenance.

Civic & utilities: Law enforcement, fire response, public records, roads, water, sewer/drainage, electric-light/power and public works.

Mining & extraction: Prospecting, drilling/blasting, ore extraction, assaying, milling, smelting, mine surveying and specialized repair/support.

9. Final Implementation Rules

Occupation is the Person's broad livelihood/work profile; it is not a character class and does not grant automatic permission to perform every task in the field.

Capabilities, equipment, workstations, materials, authority, time and location decide whether a specific task can actually be performed.

Employment terms such as seasonal, permanent, live-in, family or piecework remain in employment/agreement state rather than occupation names.

Ownership and management are relationships and authority. A Person may own a business they do not personally operate.

Civic and institutional titles such as Mayor, Sheriff, Postmaster, Judge, Foreman and Treasurer remain offices/positions where appropriate, not automatic new occupations.

Display titles may be more historically specific than the underlying profile. The UI can show Drayman, Dairy Farmer, Railroad Blacksmith or County Sheriff while the simulation uses Teamster, Farmer, Blacksmith/Farrier or Peace Officer underneath.

Do not add a new occupation solely because a historical source uses a new title. Add one only when the distinction changes capabilities, equipment, task logic, labor mobility or institutional responsibility enough to matter to simulation.

Tiering governs development order only. World generation must still depend on actual geography, population, infrastructure, institutions, capital and demand rather than forcing one of each occupation into every settlement.

10. Scope Lock Recommendation

Freeze the 1.0 occupation architecture at 112 underlying profiles. Permit additions only when later research demonstrates a material simulation distinction that cannot be represented honestly through an existing occupation plus capability, specialization, employer, office, rank or assignment. This keeps the historical surface broad while preventing occupation-specific code from ballooning.

Recommended development sequence: implement all Tier 1 profiles first, then Tier 2, then Tier 3 as the rail/market economy comes online. Tier 4 should be developed alongside the corresponding regional or infrastructure systems. Tier 5 can be completed last, but all Tier 5 profiles remain part of the intended full-release content target.

11. Source Documents - Occupation Architecture

LAND & LEDGERS - CANONICAL PROJECT BIBLE - Revision XXV - supplied project document dated 22 September 2026.

LAND & LEDGERS - TECHNICAL IMPLEMENTATION BIBLE - Revision VIII - supplied project document dated 22 September 2026.

The five-tier implementation scheme and the final 112-profile taxonomy are derived design recommendations from this review, not pre-existing Canon locks unless later incorporated into the Canon.

PART II - Population, Labor, Mortality & Workforce Calibration

This part converts the 112-occupation architecture into population-generation and labor-market calibration. Historical totals are distinguished from model-derived bridge estimates throughout. The objective is not to force every town into one demographic template; it is to provide a defensible baseline that world generation can modify by year, geography, settlement role, infrastructure, migration, household structure and local demand.

12. Core Population and Workforce Anchors

Calibration item	Value	Status	Basis
1900 South Dakota total population	401,570	DOCUMENTED	1900 U.S. Census population tables
Male population	216,164 (53.83%)	DOCUMENTED	1900 U.S. Census
Female population	185,406 (46.17%)	DOCUMENTED	1900 U.S. Census
Population age 10+	294,304	DOCUMENTED	1900 occupation/population tables
Gainfully occupied Persons age 10+	137,156	DOCUMENTED	1900 Census occupation report
Gainfully occupied / total population	34.15%	DERIVED	137,156 / 401,570
Gainfully occupied / population age 10+	46.60%	DERIVED	137,156 / 294,304
Agricultural occupations	82,857 (60.41% of workers)	DOCUMENTED + DERIVED %	1900 Census
Professional service	7,877 (5.74%)	DOCUMENTED + DERIVED %	1900 Census
Domestic & personal service	16,821 (12.26%)	DOCUMENTED + DERIVED %	1900 Census
Trade & transportation	15,274 (11.14%)	DOCUMENTED + DERIVED %	1900 Census
Manufacturing & mechanical	14,322 (10.45%)	DOCUMENTED + DERIVED %	1900 Census

> *Critical interpretation: 34.15% is a census-like gainful-occupation share of total population, not a statement that the remaining population does no work. Canon explicitly requires household and family production to consume real Person time even when no wage is paid. The game should therefore track gainful employment and productive household/family work separately.*

13. Age 0-100+ Population and Work Participation

The 1900 South Dakota age structure below is normalized across Persons with known age. The 10-44 male/female gainful-work rates use 1910 South Dakota occupation tables because they provide the needed age-by-sex breakdown; this is slightly outside the 1905 historical horizon and is therefore a late-horizon calibration rather than a direct 1870-1905 lock. The 45+ sub-bands are model splits calibrated to the 1910 combined 45+ rates. Household contribution probabilities and work-capacity factors are gameplay bridge estimates.

Age	Pop. share	Male gainful	Female gainful	M household	F household	Capacity	Game interpretation
0-4	13.778%	0.0%	0.0%	0%	0%	0.00	Dependent; no regular labor assignment.
5-9	12.987%	0.0%	0.0%	25%	35%	0.10	Chores only; schooling/play/care dominate.
10-13	9.442%	11.6%	2.1%	65%	75%	0.35	Schooling competes with chores/family work.
14-15	4.375%	34.4%	6.8%	85%	90%	0.60	Apprenticeship, farm/service work increasingly plausible.
16-20	9.852%	70.8%	25.2%	90%	92%	0.90	Near-adult work capacity; schooling/training still possible.
21-44	33.687%	95.2%	17.4%	85%	95%	1.00	Prime adult capacity; female gainful rate understates unpaid household/family production.
45-54	8.010%	92%*	13%*	85%	92%	0.95	Modeled split of documented 45+ aggregate.

Age	Pop. share	Male gainful	Female gainful	M household	F household	Capacity	Game interpretation
55-64	4.682%	82%*	8%*	75%	85%	0.80	Physical capacity begins to diverge strongly by health/occupation.
65-74	2.355%	55%*	5%*	55%	65%	0.55	No mandatory retirement; self-employment/light work remains possible.
75-84	0.752%	22%*	1.5%*	30%	35%	0.30	Mostly reduced/part-time contribution.
85-94	0.076%	5%*	0.2%*	10%	12%	0.10	Rare continued work, strongly health-dependent.
95+	0.005%	0%*	0%*	2%	2%	0.02	Terminal rare band; engine should allow ages 100+ rather than hard-cap at 100.

** 45+ male/female gainful-work sub-bands are bridge estimates chosen to reproduce the documented 1910 South Dakota combined 45+ participation rates approximately. "Household" columns mean probability of making a recurring productive household/family contribution, not a wage job. "Capacity" is an adult-equivalent work-capacity scaler before health, disability, pregnancy, injury, skill and occupation are applied.*

13.1 Sex and Gender Handling

- Use sex for the historical demographic calibration because the period Census sources report male/female categories. Social roles, identity and household relationships remain separate game data.
- Do not use the 1900 statewide 53.83% male share as the birth ratio. Generate births at an ordinary near-balanced biological ratio, then let migration, mortality and occupation-linked migration produce frontier sex imbalances.
- Occupation-specific male/female percentages in Appendix A are soft generation priors only. They never gate capability or legal eligibility by sex unless a specific historical institution/law actually requires a restriction.
- Canon explicitly rejects fixed gender penalties for domestic allocation: actual household composition, capability, availability, relationships and historically calibrated behavior determine who performs household labor.

14. How Long People Work

The game needs two different "how long" concepts: hours during a working week/season and years across a working life. Neither should be a universal 40-hour week or a mandatory retirement age. The 1900 manufacturing workweek averaged about 53 hours according to the U.S. Bureau of Labor Statistics; agriculture, domestic service, hospitality and transport were often more seasonal or irregular. The ranges below are gameplay calibration bands, not claims that every worker in the category worked exactly these hours.

Work family	Baseline calibration	Implementation note
Clerical / professional / office	48-58 h	Generally steady; some self-employed professionals extend hours.
Skilled trades / manufacturing	50-60 h	Use 53 h around 1900 as a strong manufacturing anchor; overtime/seasonal demand can raise it.
Retail / hospitality	55-70 h presence	Long opening/presence hours; active task time can be lower than establishment presence.
Agriculture - ordinary season	45-60 h active	Task-driven; strongly dependent on daylight, weather and enterprise mix.
Agriculture - planting/harvest peak	70-90 h active	Bridge range for short peaks; fatigue and weather should matter.
Agriculture - winter/low season	30-45 h active	Livestock/maintenance continue; field-work demand falls.
Transport / railroad / river	55-70 h on-duty	Irregular schedules, trips, waiting and duty periods should replace modern shift assumptions.
Domestic live-in service	60-80 h availability	Separate availability/presence from active task time to avoid treating every minute as full-intensity labor.
Mining / logging / heavy construction	50-60 h	Weather, shutdowns and project seasonality can reduce annual utilization despite long working weeks.

Career entry and "soft exit" ages are listed for every occupation in Appendix A. Soft exit means the age band where full-time participation should begin falling materially; it is not retirement, death, or a hard prohibition. Owners, farmers, merchants, clergy and professionals may continue much later if health permits.

15. Mortality, Injury and Occupation Risk

Do not assign a fixed life expectancy to an occupation. A Laborer should die younger on average than a Bookkeeper only because the Laborer accumulates more accident exposure, physical strain and possibly chronic damage - not because “Laborer” contains a scripted death age. Baseline mortality should come first from age, sex, calendar period and health; occupation adds a smaller stochastic work-risk component.

Mortality anchor	Value	Scope
Survive birth to age 20 - white male	76.4%	1900-1902 death-registration States
Survive birth to age 20 - white female	79.0%	1900-1902 death-registration States
Survive birth to age 65 - white male	39.2%	1900-1902 death-registration States
Survive birth to age 65 - white female	43.8%	1900-1902 death-registration States
Median age at death - white male	57.2 years	1900-1902 death-registration States
Median age at death - white female	60.6 years	1900-1902 death-registration States

These life-table anchors are not South Dakota-specific. The federal 1900-1902 life tables cover the contemporary death-registration area (10 States plus the District of Columbia). They are best used as national-period mortality calibration, then modified by Land & Ledgers geography, disease environment, childbirth, accidents, health and care access.

Hazard class	Relative work-risk multiplier	Meaning
H0 - Sheltered	0.25x	Very low occupational accident/exposure component: clerical, bookkeeping, drafting.
H1 - Low	0.50x	Light commercial/professional/service work.
H2 - Moderate	1.00x	Ordinary workshop/service exposure; baseline work-risk component.
H3 - Elevated	2.00x	Heavy manual work, animal/vehicle exposure, machinery, construction or notable infection/chemical exposure.
H4 - High	4.00x	Rail/river operations, heavy machinery, survey/frontier exposure, hazardous industrial work.
H5 - Extreme	8.00x	Logging, blasting, mining, switching/brakeman work, linemen and similarly dangerous work.

Strain	Band	Meaning
S0	Sedentary	Little physical load; work capacity declines comparatively slowly with age.
S1	Light	Standing/walking/light manual work.
S2	Moderate	Regular manual effort but not sustained heavy exertion.
S3	Heavy	Frequent lifting, awkward work, animals, machinery or prolonged physical demand.
S4	Very heavy	Sustained heavy labor or physically punishing/unsafe working conditions.

The H multipliers apply only to a calibrated occupational-accident/exposure component, not to total all-cause mortality. They are bridge parameters, not measured 1880 Dakota fatality ratios. BLS historical summaries show that early-20th-century industrial work was dramatically more dangerous than modern employment, supporting a separate occupational hazard layer. Injury/disability should use the same ordering but occur much more often than fatal events.

16. Baseline Occupation Weight Model - 100.000% of Gainful Workers

The weight denominator is the gainfully occupied worker pool, not total population. In the 1900 South Dakota anchor, the broad Census classes sum to 100% of 137,156 gainfully occupied Persons. The 112 detailed percentages are model-derived bridge weights normalized inside those broad classes. They are intended as a mature statewide/rural-service-district prior, not as fixed quotas for a town core.

Broad class	Worker share	Status
Agriculture	60.410%	1900 Census anchor
Professional service	5.740%	1900 Census anchor
Domestic & personal service	12.260%	1900 Census anchor
Trade & transportation	11.140%	1900 Census anchor

Broad class	Worker share	Status
Manufacturing & mechanical	10.450%	1900 Census anchor
TOTAL	100.000%	Normalization target

#	Occupation	Tier	Base worker wt.	Male prior*	Risk
1	Farmer	1	40.120%	93.3%	H3/S3
3	Farm Hand	1	19.500%	90.0%	H3/S4
8	Laborer	1	3.819%	95.0%	H3/S4
51	Domestic Servant	2	3.819%	15.0%	H2/S3
70	Teacher	1	3.553%	30.0%	H1/S1
9	Teamster	1	2.763%	99.0%	H3/S3
34	Merchant	1	2.438%	92.0%	H1/S1
14	Carpenter	1	1.952%	99.0%	H3/S3
35	Clerk	2	1.950%	80.0%	H0/S0
52	Laundress	2	1.273%	2.0%	H2/S4
30	Dressmaker / Seamstress	1	1.153%	5.0%	H1/S1
13	Blacksmith / Farrier	1	0.976%	99.0%	H3/S3
90	Track Worker	3	0.975%	99.0%	H5/S4
40	Hospitality Keeper	1	0.700%	60.0%	H2/S2
42	Cook	2	0.700%	55.0%	H2/S3
71	Clergy	2	0.611%	98.0%	H1/S1
44	Housekeeper	2	0.573%	10.0%	H2/S3
36	Bookkeeper / Accountant	3	0.488%	85.0%	H0/S0
53	Logger	4	0.444%	99.0%	H5/S4
38	Agent	3	0.406%	95.0%	H1/S1

The Appendix contains all 112 weights. Displayed weights are rounded by a largest-remainder method so the printed 112 rows sum to exactly 100.000%. Male priors are soft bridge estimates calibrated so the weighted occupational pool is approximately 84.4% male, matching the strongly male-skewed 1900 Census gainful-worker structure; they are not access restrictions.

17. Settlement, Infrastructure and Historical-Phase Modifiers

Never apply the 112 percentages directly to every settlement. Generate or solve the rural catchment and built-up settlement core separately. Apply prerequisite gates and local multipliers, then renormalize the surviving weights. The following multipliers are starting calibration guidance rather than historical constants.

Context	Starting modifier	Interpretation
No prerequisite infrastructure	Set occupation weight to 0	No rail jobs without rail; no riverboat jobs without navigable service/landing; no mining jobs without mines; no telephone/electric jobs without networks.
1870-72 sparse frontier	Rail 0; telephone/electric 0; banking ~0.25x; core farm/teamster/smith/merchant/service weights rise after renormalization	Matches Canon opening: wagon/river logistics, thin finance, promoted/under-construction networks.
1878-87 rail & grain boom	Rail profiles 3-5x; grain/elevator 2-4x; teamster 1.2-1.5x; merchant/clerk 1.3-1.8x	Use only where track/service and grain market actually exist.
County seat / administrative center	Lawyer, Clerk, Banker, Agent, Peace Officer, Physician, print 2-4x	Institutional role drives specialization more than raw population alone.
Mining settlement	Mining profiles 20-60x; town-core agriculture 0.1-0.3x; teamster/hospitality/saloon 2-4x	Rural catchment, if any, remains separate.
River node	River profiles 15-40x; teamster/agent/hospitality 1.5-3x	Requires real river/landing/service context.
Mature regional center, 1896-1905	Rare/specialist profiles 2-5x when supporting institutions exist	Photographer, dentist, architect, telephone/electrical and other specialists become more plausible.
Rural service district	~1.0x baseline	Closest use case for the statewide/rural-service weight prior.

17.1 Recommended Generation Formula

- 1. Generate Persons and households first from age/sex/migration/fertility/mortality rules.

- 2. Determine gainful-work participation from age, sex, household situation, schooling, health and local opportunity; separately allocate unpaid household/family production.
- 3. For each eligible worker, compute adjusted occupation weight: base_weight x phase_modifier x settlement_role_modifier x infrastructure_gate x service_demand_modifier x individual_fit.
- 4. Renormalize adjusted weights across occupations that are actually possible in that world.
- 5. Sample stochastically or solve against service demand; do not deterministically round percentages in small settlements.
- 6. Assign employer, employment agreement, current tasks, office/rank and display title after the underlying occupation is chosen.

For a 2,000-worker population, a 0.010% occupation weight means an expected value of only 0.2 workers before local modifiers. That is intentional: rare professions should appear probabilistically or when a supporting institution creates real demand, not because every city is forced to contain one.

18. Occupation Sex Priors, Migration and Household Work

The occupation matrix includes a male-generation prior for each job because the 1870-1905 labor market was strongly sex-segregated. These priors should influence who appears in a generated historical population but must never be coded as innate capability penalties. A female blacksmith or male domestic servant remains mechanically possible when the Person, household, employer and social circumstances produce it.

- Use occupation sex priors mainly when generating migrants, job applicants and initial historical populations.
- Use household labor allocation separately. Census “gainful occupation” categories miss much productive domestic/family work that Canon requires the simulation to represent.
- Let migration create local imbalance. Railroad construction, mining camps and frontier labor inflows can skew male; schools, domestic service, family settlement and maturing communities change the balance over time.
- Do not make sex priors constant across the entire 1870-1905 horizon. Treat Appendix A values as a baseline that later research/scenario data can override.

19. Quick Validation Examples

World type	Expected behavior
55-95 Person spring-1870 settlement	Do not “fill the roster.” Many occupations have zero local workers. The settlement may rely on one Merchant/Postmaster, one Blacksmith/Farrier, a Teacher, a Hospitality Keeper and services from the wider rural catchment.
Rail/grain town after service opens	Add Station Agent, Track Workers and train occupations only when service exists; Elevator Operator and Grain Dealer expand as actual grain throughput supports them; Teamsters remain important for farm-to-depot and local drayage.
County seat	Professional/civic work rises: Clerk, Lawyer, Banker, Physician, Agent, print occupations and Peace Officer. Offices remain layered on occupations.
Mining town	Mining occupations dominate the town-core worker mix; Farm/Farm Hand weights should not be imported from the statewide prior into the built-up core.
City of several thousand	Use service-demand and institutional thresholds so 0.005-0.05% rare jobs can appear without being universal. Sample rare professions over the broader urban/regional labor market rather than one block.

20. Appendix A - Full 112-Occupation Workforce Calibration Matrix

Base Weight is percent of the gainfully occupied worker pool before local modifiers. M Prior is a soft male-generation prior; F Prior is its complement. Entry is the typical age range for beginning recognizable paid/primary work in the occupation; Soft Exit is the age range where full-time participation should begin to fall materially. H is occupational hazard and S is physical strain. None of these fields is a hard rule.

#	Occupation	Tier	Weight	M prior	F prior	Entry	Soft exit	Hazard / strain
1	Farmer	1	40.120%	93.3%	6.7%	18-25	65-75	H3/S3
2	Rancher	4	0.250%	98.0%	2.0%	18-25	60-70	H3/S3
3	Farm Hand	1	19.500%	90.0%	10.0%	12-16	55-65	H3/S4
4	Stock Hand / Range Hand	4	0.180%	99.0%	1.0%	14-18	50-60	H4/S4
5	Agricultural Machinery Operator	3	0.200%	99.0%	1.0%	16-20	55-65	H4/S3
6	Horse Trainer	4	0.030%	95.0%	5.0%	16-20	55-65	H3/S3
7	Stable Hand	2	0.130%	95.0%	5.0%	12-16	55-65	H3/S3
8	Laborer	1	3.819%	95.0%	5.0%	14-17	50-60	H3/S4
9	Teamster	1	2.763%	99.0%	1.0%	15-18	55-65	H3/S3
10	Stagecoach Driver	2	0.081%	99.0%	1.0%	18-21	55-65	H4/S3
11	Messenger	3	0.130%	90.0%	10.0%	10-14	50-65	H2/S2
12	Mail Carrier	2	0.195%	98.0%	2.0%	18-21	60-70	H3/S3
13	Blacksmith / Farrier	1	0.976%	99.0%	1.0%	14-18	60-70	H3/S3
14	Carpenter	1	1.952%	99.0%	1.0%	14-18	60-70	H3/S3
15	Cabinetmaker	3	0.213%	99.0%	1.0%	14-18	60-70	H2/S2
16	Mason	2	0.355%	99.0%	1.0%	14-18	55-65	H3/S4
17	Stonecutter	4	0.071%	99.0%	1.0%	14-18	50-60	H4/S4
18	Plasterer	3	0.142%	99.0%	1.0%	14-18	55-65	H2/S3
19	Painter / Glazier	2	0.319%	98.0%	2.0%	14-18	55-65	H3/S3
20	Tinsmith	2	0.213%	99.0%	1.0%	14-18	60-70	H2/S2
21	Plumber	4	0.053%	99.0%	1.0%	16-20	60-70	H2/S3
22	Mechanic	3	0.266%	99.0%	1.0%	16-20	55-65	H3/S3
23	Machinist	4	0.142%	99.0%	1.0%	16-20	55-65	H3/S2
24	Millwright	4	0.089%	99.0%	1.0%	16-20	55-65	H4/S3
25	Boilermaker	4	0.036%	99.0%	1.0%	16-20	50-60	H4/S4
26	Wagonmaker / Wheelwright	1	0.319%	99.0%	1.0%	14-18	60-70	H2/S3
27	Saddler / Harness Maker	1	0.319%	99.0%	1.0%	14-18	60-70	H2/S2
28	Shoemaker	2	0.390%	98.0%	2.0%	12-16	60-70	H1/S2
29	Tailor	2	0.213%	90.0%	10.0%	14-18	60-70	H1/S1
30	Dressmaker / Seamstress	1	1.153%	5.0%	95.0%	12-16	55-70	H1/S1
31	Jeweler / Watchmaker	5	0.071%	98.0%	2.0%	14-18	65-75	H1/S1
32	Gunsmith	5	0.018%	99.0%	1.0%	16-20	60-70	H2/S2
33	Specialty Artisan	5	0.213%	80.0%	20.0%	14-18	55-70	H2/S2
34	Merchant	1	2.438%	92.0%	8.0%	18-25	65-75	H1/S1
35	Clerk	2	1.950%	80.0%	20.0%	14-18	65-75	H0/S0
36	Bookkeeper / Accountant	3	0.488%	85.0%	15.0%	16-20	65-75	H0/S0
37	Banker	3	0.130%	99.0%	1.0%	22-30	65-80	H0/S0
38	Agent	3	0.406%	95.0%	5.0%	18-25	65-75	H1/S1
39	Grain Dealer	3	0.406%	99.0%	1.0%	18-25	65-75	H2/S1
40	Hospitality Keeper	1	0.700%	60.0%	40.0%	18-25	65-75	H2/S2
41	Saloon Keeper	2	0.318%	97.0%	3.0%	18-25	60-70	H2/S2
42	Cook	2	0.700%	55.0%	45.0%	14-18	55-65	H2/S3
43	Server	2	0.318%	45.0%	55.0%	14-18	55-65	H2/S3
44	Housekeeper	2	0.573%	10.0%	90.0%	12-16	55-65	H2/S3

#	Occupation	Tier	Weight	M prior	F prior	Entry	Soft exit	Hazard / strain
45	Bartender	2	0.280%	95.0%	5.0%	16-20	55-65	H2/S2
46	Gambling Dealer	4	0.025%	98.0%	2.0%	18-25	55-65	H2/S1
47	Musician	4	0.051%	85.0%	15.0%	14-20	60-75	H1/S1
48	Entertainer	5	0.013%	60.0%	40.0%	14-20	50-65	H2/S2
49	Sex Worker	5	0.025%	5.0%	95.0%	16-20	35-50	H3/S2
50	Barber	2	0.153%	99.0%	1.0%	14-18	65-75	H1/S1
51	Domestic Servant	2	3.819%	15.0%	85.0%	12-16	50-60	H2/S3
52	Laundress	2	1.273%	2.0%	98.0%	12-16	50-60	H2/S4
53	Logger	4	0.444%	99.0%	1.0%	14-18	45-60	H5/S4
54	Sawyer	3	0.213%	99.0%	1.0%	14-18	55-65	H4/S3
55	Miller	2	0.355%	99.0%	1.0%	14-18	60-70	H3/S3
56	Dairy Processor	3	0.089%	40.0%	60.0%	14-18	55-65	H2/S3
57	Brewer	4	0.027%	99.0%	1.0%	16-20	60-70	H2/S3
58	Baker	2	0.284%	85.0%	15.0%	14-18	55-65	H2/S3
59	Butcher	2	0.355%	99.0%	1.0%	14-18	55-65	H3/S3
60	Elevator Operator	3	0.213%	99.0%	1.0%	16-20	60-70	H3/S3
61	Grain Weigher / Inspector	3	0.089%	99.0%	1.0%	16-20	60-70	H2/S2
62	Physician	2	0.278%	98.0%	2.0%	24-30	65-80	H2/S1
63	Nurse	4	0.089%	10.0%	90.0%	18-25	55-70	H2/S3
64	Midwife	2	0.111%	1.0%	99.0%	18-25	55-70	H2/S3
65	Dentist	5	0.033%	99.0%	1.0%	22-30	65-75	H1/S1
66	Veterinarian	4	0.044%	99.0%	1.0%	20-28	60-70	H3/S3
67	Druggist / Pharmacist	2	0.293%	95.0%	5.0%	18-25	65-75	H1/S1
68	Medical Attendant	4	0.038%	35.0%	65.0%	16-22	55-65	H2/S3
69	Lawyer	3	0.222%	99.0%	1.0%	22-30	65-80	H0/S0
70	Teacher	1	3.553%	30.0%	70.0%	17-22	55-70	H1/S1
71	Clergy	2	0.611%	98.0%	2.0%	22-30	65-80	H1/S1
72	Undertaker	3	0.051%	98.0%	2.0%	18-25	65-75	H2/S2
73	Editor / Publisher	3	0.133%	95.0%	5.0%	18-25	65-75	H1/S1
74	Reporter	4	0.033%	90.0%	10.0%	16-22	60-70	H2/S2
75	Compositor	3	0.142%	90.0%	10.0%	14-18	55-65	H2/S2
76	Printer	3	0.213%	95.0%	5.0%	14-18	55-65	H3/S3
77	Photographer	5	0.028%	95.0%	5.0%	16-22	60-70	H2/S2
78	Artist	5	0.011%	80.0%	20.0%	16-22	65-75	H1/S1
79	Professional Engineer	4	0.044%	99.0%	1.0%	22-30	65-75	H2/S1
80	Architect	5	0.011%	99.0%	1.0%	22-30	65-75	H1/S0
81	Draftsman	4	0.033%	99.0%	1.0%	16-22	65-75	H0/S0
82	Surveyor	2	0.200%	99.0%	1.0%	18-24	60-70	H3/S3
83	Stationary Engineer	3	0.106%	99.0%	1.0%	18-22	55-65	H4/S3
84	Peace Officer	2	0.178%	99.0%	1.0%	18-25	55-65	H3/S3
85	Detective	5	0.006%	99.0%	1.0%	20-30	60-70	H2/S2
86	Guard	3	0.102%	99.0%	1.0%	18-25	60-70	H2/S2
87	Firefighter	4	0.022%	99.0%	1.0%	18-25	50-60	H5/S4
88	Scout / Guide	5	0.011%	99.0%	1.0%	16-22	50-65	H4/S4
89	Station Agent	3	0.130%	99.0%	1.0%	18-25	60-70	H2/S1
90	Track Worker	3	0.975%	99.0%	1.0%	14-18	50-60	H5/S4
91	Brakeman	3	0.130%	99.0%	1.0%	18-22	45-60	H5/S4
92	Conductor	3	0.081%	99.0%	1.0%	20-28	55-65	H4/S3
93	Locomotive Engineer	3	0.081%	99.0%	1.0%	20-28	55-65	H4/S2
94	Train Dispatcher	4	0.024%	99.0%	1.0%	20-28	60-70	H1/S1
95	Switchman	4	0.049%	99.0%	1.0%	18-22	50-60	H5/S4
96	Riverboat Officer	4	0.016%	99.0%	1.0%	20-30	60-70	H4/S2
97	Riverboat Pilot	4	0.010%	99.0%	1.0%	20-30	60-70	H4/S2

#	Occupation	Tier	Weight	M prior	F prior	Entry	Soft exit	Hazard / strain
98	Riverboat Engineer	4	0.013%	99.0%	1.0%	20-28	55-65	H4/S3
99	Steam Fireman	3	0.130%	99.0%	1.0%	16-20	45-60	H5/S4
100	Deckhand	4	0.033%	99.0%	1.0%	14-18	45-60	H5/S4
101	Ferryman	2	0.049%	99.0%	1.0%	16-22	55-65	H4/S3
102	Telegraph Operator	3	0.130%	90.0%	10.0%	16-22	60-70	H1/S0
103	Telephone Operator	5	0.008%	5.0%	95.0%	16-22	45-60	H0/S0
104	Lineman	3	0.106%	99.0%	1.0%	16-22	45-60	H5/S4
105	Electrician	5	0.018%	99.0%	1.0%	18-24	55-65	H4/S3
106	Miner	4	0.177%	99.0%	1.0%	14-18	45-55	H5/S4
107	Driller	4	0.027%	99.0%	1.0%	16-20	45-55	H5/S4
108	Powderman	4	0.018%	99.0%	1.0%	18-25	45-55	H5/S3
109	Assayer / Metallurgist	5	0.009%	99.0%	1.0%	20-28	60-70	H3/S1
110	Ore-Mill Operator	4	0.027%	99.0%	1.0%	16-20	50-60	H4/S4
111	Smelter Worker	4	0.018%	99.0%	1.0%	16-20	45-60	H5/S4
112	Soldier	4	0.089%	99.0%	1.0%	18-25	40-60	H4/S4

20.1 Appendix Interpretation

Field	Meaning
Weight	Percent of the baseline gainfully occupied worker pool; printed 112-row total = 100.000%. Apply world modifiers and renormalize.
M/F prior	Soft historical generation prior only. Never a capability or eligibility gate.
Entry	Typical beginning of recognizable occupation work; apprenticeship/family assistance can precede it.
Soft exit	Age band where full-time participation should decline; no mandatory retirement or scripted death age.
H0-H5	Occupational accident/exposure hazard class; affects only the work-risk component of mortality/injury.
S0-S4	Physical strain class; affects fatigue, injury recovery, aging-related work capacity and likely exit from heavy work.

21. Evidence Labels, Limitations and Sources

DOCUMENTED: Directly reported by the cited Census, federal historical statistical source, Canon or Technical Bible.

DERIVED: Arithmetic transformation of documented numbers, such as a percentage or normalized age share.

BRIDGE ESTIMATE: A game-calibration value designed to connect historical aggregates to a simulation field not directly reported by the source.

SCENARIO OVERRIDE: A value expected to change materially by settlement role, geography, infrastructure, year or authored scenario data.

Source	Use in this report
LAND & LEDGERS - Canonical Project Bible, Revision XXV (22 Sep 2026).	Primary project authority for 1870 opening, settlement envelope, transport, household labor and historical-system doctrine.
LAND & LEDGERS - Technical Implementation Bible, Revision VIII (22 Sep 2026).	Technical planning reference for farm-generation inputs, EmploymentRelationship authority and occupation profiles as data rather than class gates.
U.S. Census Bureau, Occupations at the Twelfth Census (1900), Special Reports, published 1904.	Primary anchor for South Dakota gainfully occupied workers and broad occupation classes; occupation inquiry covered gainfully employed Persons age 10+.
U.S. Census Bureau, Twelfth Census population tables (1900), South Dakota.	Primary anchor for total population, sex composition and age structure.
U.S. Census Bureau, Abstract of the Thirteenth Census - Supplement for South Dakota (1910).	Near-horizon anchor for gainful occupation participation by age and sex; used cautiously because 1910 is five years beyond the authored historical horizon.
National Center for Health Statistics, U.S. life-table historical series, 1900-1902 death-registration States.	Period mortality/survival anchor. Not South Dakota-specific; geographic coverage was 10 registration States plus the District of Columbia.
U.S. Bureau of Labor Statistics, American Labor in the 20th Century.	Historical workweek and occupational-safety context; reports average manufacturing workweek of about 53 hours in 1900.

Public source links

https://www.census.gov/library/publications/1904/dec/occupations.html
https://www2.census.gov/library/publications/decennial/1900/volume-2/volume-2-p2.pdf
https://www2.census.gov/library/publications/decennial/1910/abstract/supplement-south-dakota.pdf
https://www.cdc.gov/nchs/data/lifetables/life92_2.pdf
https://www.bls.gov/opub/mlr/cwc/american-labor-in-the-20th-century.pdf

22. Revised Scope Lock

Keep the full-release occupation architecture at 112 underlying profiles. Add no new profile merely because research finds a new historical job title. Add one only if the new role materially changes capability requirements, equipment, task logic, labor mobility, institutional responsibility or occupational risk in a way that cannot be represented through specialization, employer, office, rank, agreement or assignment.

Treat the demographic and workforce numbers in Part II as versioned calibration data rather than immutable Canon. The broad 1900 Census anchors are historical facts; the detailed weights, sex priors, work-capacity factors, age splits and hazard classes are explicitly designed to be tuned through autonomous playtests, scenario validation and future primary-source research without changing the underlying occupation ontology.

System	Final recommendation	Meaning
Occupation profiles	112	Scope-locked ontology
Detailed baseline weights	112; sum = 100.000%	Worker-pool prior, not town quotas
Development tiers	5	Implementation priority, not social rank
Age model	0-100+	12 practical bands + single-year ages in Person state
Labor concepts	2	Gainful occupation + productive household/family work
Risk layers	H0-H5 + S0-S4	Stochastic occupation risk/strain, not fixed death ages